

PRAJUAL PILLAI

ML Engineer

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EXPERIENCE

ML Engineer II

Nissan Digital

September 2023 - Ongoing

Trivandrum, India

Carflow Prediction: Nissan NNA

Jan, 2025 – Mar, 2025

- Developed a forecasting tool to predict the car sales for each month
- Reduced the average **6 month mape** from **32% to 12%**
- Forecast uses multiple feature engineering and forecasting techniques to find the best feature combination and tune the model

Powertrain Mix: NML Japan

Oct, 2024 – Jan, 2025

- Developed a regression-based market share simulation tool for the USA (two segments) and France (one segment) regions
- Developed a **random choice modeling** based solution for NML France
- Developed and implemented variable forecasts using ARIMA, exponential smoothing, Prophet, and mean averaging techniques.
- Developed charts(bar & line) to show variable impact on the market mix

MMM: Nissan Philippines

Dec, 2023 – Sep, 2024

- Developed a custom **constrained regression** based **Market Mix Model** for Nissan Altima, Terra and Navara with an avg N+1 **mape** of **10%**
- Performed intensive EDA to analyze the spend & sales data for each car.
- Used **geometric adstocking** to determine the actual impact of spending.
- Modeled **saturation** curves with **Hill function** to find key points.

ML Engineer

Seance Inc.

July 2021 – September 2023

Pune, India

Client: Futureproof

July, 2023 – July, 2023

- Developed a CI/CD pipeline model training and versioning using **DVC**
- Set up the architecture for integrating **DVC** with **VertexAI**

Client: Seance

March, 2023 – April, 2023

- Built segmentation & NER model from scratch using **YOLOv8** & **spacy3.1**
- Extracted **9** entities from driving licenses with an accuracy of **93.1%**
- Extracted photo and signature with accuracies of **95.7%** and **78.3%**

Client: Healthians

September, 2022 – February, 2023

- Developed collaborative filter based recommendation system using **6TB** of data to recommend medical tests to clients
- Trained, tuned & deployed an **80%** accurate model in **AWS Sagemaker**
- Deployment of the same also done using **AWS Lambda** & **AWS ECR**
- Setup **AWS API Gateway** as the trigger for the Lambda deployment

Client: Flipkart

January – September, 2022

- Worked with Google team to migrate **3PB** of on-prem data to GCS
- Mapped bucket & object policy from **AWS** to **GCS** at scale
- Deployed the **multi-threaded** python code on **GKE** for data migration
- Performed data verification & analysis of the copied objects at **scale**

EDUCATION

M.Tech, IIT Bombay

Manufacturing

Performance- 8.79

2021

B.Tech, Kerala University

Mechanical

Performance- 7.45

2017

Intermediate, St. Fidelis College

I.C.S.E

Performance- 81.3

2013

ACHIEVEMENTS

Secured **99.22** percentile in Graduate Aptitude Test in Engineering(GATE-19) among **1.67 lakh** candidates

CERTIFICATES

GCP: Professional ML Engineer, Associate Cloud Engineer

AWS: AWS Certified Cloud Practitioner

Coursera: Applied Data Science Capstone, Neural Networks and Deep Learning

TECHNICAL SKILLS

Programming: Python, Java, SQL

AWS: S3, Sagemaker, ECR, Lambda, Glue

GCP: GCS, BigQuery, Pub/Sub, Spanner, Cloud Functions, Dataflow, VertexAI, Cloud Scheduler, **Cloud Composer**, **DocAI**

ML Frameworks and Libraries: Tensorflow, PyTorch, Spacy, BERT, LLM, RAG, Langchain

EXTRA CURRICULAR

- Represented Mechanical Dept. in the PG inter-department volleyball tournament at IITB.

BLOGS

- Custom Recommendation System
- Performing NER using spacy3.0